Generative Models in Finance — Coursework

Spring 2026 · Cristopher Salvi · Imperial College London

Deadline: 1 April 2026 · **Groups of 2** · **Weight:** 100%

Deliverables. Submit a single `.zip` file to c.salvi@imperial.ac.uk by the deadline containing:

- One report (PDF) covering Parts I and III.
- A Jupyter notebook (`.ipynb`) and any auxiliary files for Part II.

Part I: Mathematics — From Denoising to Flow Matching

*The following papers are **required reading** for this part. The problems below guide you through their key results.*

[H] A. Hyvärinen. *Estimation of Non-Normalized Statistical Models by Score Matching.* JMLR, 2005.

[V] P. Vincent. *A Connection Between Score Matching and Denoising Autoencoders.* Neural Computation, 2011.

[SE] Y. Song and S. Ermon. *Generative Modeling by Estimating Gradients of the Data Distribution.* NeurIPS, 2019.

[Ho] J. Ho, A. Jain, and P. Abbeel. *Denoising Diffusion Probabilistic Models.* NeurIPS, 2020.

[S+] Y. Song et al. *Score-Based Generative Modeling through SDEs.* ICLR, 2021.

[L+] Y. Lipman et al. *Flow Matching for Generative Modeling.* ICLR, 2023.

[A+] M.S. Albergo, N.M. Boffi, and E. Vanden-Eijnden. *Stochastic Interpolants: A Unifying Framework for Flows and Diffusions.* arXiv:2303.08797, 2023.

Throughout, let $z \sim p_{\text{data}}$ on $\mathbb{R}^d$, $\epsilon \sim \mathcal{N}(0, I_d)$ independent of z, and $x = \alpha_t z + \beta_t \epsilon$ for noise schedulers α_t, β_t with $\alpha_0 = \beta_1 = 0$, $\alpha_1 = \beta_0 = 1$. The conditional distribution is $p_t(x \mid z) = \mathcal{N}(\alpha_t z, \beta_t^2 I_d)$.

Problem 1: Implicit score matching (Hyvärinen, 2005)**.** *The score $\nabla \log p(x)$ of a distribution p is unknown, yet we can still set up a tractable objective to learn it. Here p denotes a generic density on $\mathbb{R}^d$ (not necessarily p_t).*

(a) Let $s_\theta : \mathbb{R}^d \to \mathbb{R}^d$ be a parametric model. Write down the **explicit score matching** objective:

$$J_{\text{ESM}}(\theta) = \frac{1}{2}\mathbb{E}_{x \sim p}\big[\|s_\theta(x) - \nabla \log p(x)\|^2\big].$$

Expand the square and identify the term that depends on θ but involves the unknown $\nabla \log p$.

(b) Using integration by parts, show that for each component i:

$$\mathbb{E}_{x \sim p}\big[s_{\theta,i}(x)\, \partial_{x_i} \log p(x)\big] = -\mathbb{E}_{x \sim p}\big[\partial_{x_i} s_{\theta,i}(x)\big],$$

provided $p(x)\, s_{\theta,i}(x) \to 0$ as $\|x\| \to \infty$.

(c) Conclude that, up to a constant independent of θ:

$$\boxed{J_{\text{ESM}}(\theta) = \mathbb{E}_{x\sim p}\left[\sum_{i=1}^{d} \partial_{x_i} s_{\theta,i}(x) + \frac{1}{2}\|s_\theta(x)\|^2\right] + C.}$$

This is the **implicit score matching** (ISM) objective: it only requires samples from p, not $\nabla \log p$ itself.

Problem 2: Tweedie's formula. *The posterior mean $\mathbb{E}[z \mid x]$ can be expressed in terms of the marginal score $\nabla \log p_t(x)$.*

(a) Show that $\nabla_x p_t(x \mid z) = -\dfrac{x - \alpha_t z}{\beta_t^2}\, p_t(x \mid z)$.

(b) Using $p_t(x) = \int p_t(x \mid z)\, p_{\text{data}}(z)\, \mathrm{d}z$, show that

$$\nabla \log p_t(x) = -\frac{x - \alpha_t \mathbb{E}[z \mid x]}{\beta_t^2}.$$

(c) Deduce **Tweedie's formula**:

$$\boxed{\mathbb{E}[z \mid x] = \frac{1}{\alpha_t}\big(x + \beta_t^2 \nabla \log p_t(x)\big).}$$

Problem 3: Denoising score matching. *We prove that regressing a network $s_t^\theta(x)$ against the conditional score $\nabla \log p_t(x \mid z)$ is equivalent to matching the (intractable) marginal score $\nabla \log p_t(x)$.*

(a) Write down the **marginal** and **conditional** score matching losses:

$$\mathcal{L}_{\text{SM}}(\theta) = \mathbb{E}_{t,\, x\sim p_t}\big[\|s_t^\theta(x) - \nabla \log p_t(x)\|^2\big], \qquad \mathcal{L}_{\text{CSM}}(\theta) = \mathbb{E}_{t,\, z,\, x\sim p_t(\cdot|z)}\big[\|s_t^\theta(x) - \nabla \log p_t(x \mid z)\|^2\big].$$

Expand $\mathcal{L}_{\text{SM}}$ as $\mathbb{E}[\|s_t^\theta\|^2] - 2\mathbb{E}[(s_t^\theta)^T \nabla \log p_t] + C_1$, where C_1 is independent of θ.

(b) Show that the cross term satisfies

$$\mathbb{E}_{t,\, x\sim p_t}\big[(s_t^\theta(x))^T \nabla \log p_t(x)\big] = \mathbb{E}_{t,\, z,\, x\sim p_t(\cdot|z)}\big[(s_t^\theta(x))^T \nabla \log p_t(x \mid z)\big].$$

(c) Conclude that $\mathcal{L}_{\text{SM}}(\theta) = \mathcal{L}_{\text{CSM}}(\theta) + C$ for a constant C independent of θ.

Problem 4: From score matching to DDPM. *We show that the denoising diffusion (DDPM) ϵ-prediction loss is a reparametrisation of conditional score matching.*

(a) Using Problem 2(a), show that $\nabla \log p_t(x \mid z) = -\epsilon/\beta_t$ when $x = \alpha_t z + \beta_t \epsilon$.

(b) Define the **noise predictor** $\epsilon_t^\theta(x) := -\beta_t\, s_t^\theta(x)$. Substitute into $\mathcal{L}_{\text{CSM}}$ to show

$$\mathcal{L}_{\text{CSM}}(\theta) = \frac{1}{\beta_t^2}\, \mathbb{E}_{t,\, z,\, \epsilon}\big[\|\epsilon_t^\theta(\alpha_t z + \beta_t \epsilon) - \epsilon\|^2\big].$$

(c) Explain why minimising $\mathcal{L}_{\text{CSM}}$ is equivalent to minimising the **DDPM loss** $\mathcal{L}_{\text{DDPM}}(\theta) = \mathbb{E}_{t,z,\epsilon}[\|\epsilon_t^\theta(\alpha_t z + \beta_t \epsilon) - \epsilon\|^2]$. What is the role of the $1/\beta_t^2$ weighting?

Problem 5: Equivalence with flow matching. *We show that, for Gaussian paths, a trained score network can be converted into a flow matching vector field, and vice versa.*

(a) Recall the conditional vector field $u_t^{\text{target}}(x \mid z) = (\dot{\alpha}_t - \frac{\dot{\beta}_t}{\beta_t}\alpha_t)z + \frac{\dot{\beta}_t}{\beta_t}x$ from the lectures. Show that it can be rewritten as

$$u_t^{\text{target}}(x \mid z) = \frac{\dot{\alpha}_t}{\alpha_t}x + \left(\frac{\dot{\alpha}_t\beta_t^2}{\alpha_t} - \dot{\beta}_t\beta_t\right)\nabla \log p_t(x \mid z).$$

(b) Using the identity from Problem 3(b) (replacing scores by vector fields via the same marginalisation structure), argue that the same conversion holds at the marginal level:

$$\boxed{u_t^{\text{target}}(x) = \frac{\dot{\alpha}_t}{\alpha_t}x + \left(\frac{\dot{\alpha}_t\beta_t^2}{\alpha_t} - \dot{\beta}_t\beta_t\right)\nabla \log p_t(x).}$$

(c) Specialise to the CondOT path ($\alpha_t = t,\ \beta_t = 1 - t$). Write out $u_t^{\text{target}}(x)$ explicitly in terms of x and $\nabla \log p_t(x)$, and verify that $u_t^{\text{target}}(x \mid z) = z - \epsilon$ when $x = tz + (1-t)\epsilon$.

Problem 6: The probability flow ODE. *We derive a deterministic ODE whose trajectories have the same marginal distributions as a given SDE — this is the probability flow ODE (Song et al., 2021).*

(a) Recall the **Fokker–Planck equation** from the lectures: the SDE $\mathrm{d}X_t = u_t(X_t)\,\mathrm{d}t + \sigma_t\,\mathrm{d}W_t$ with $X_0 \sim p_0$ satisfies $X_t \sim p_t$ if and only if

$$\partial_t p_t = -\mathrm{div}(p_t\, u_t) + \tfrac{\sigma_t^2}{2}\,\Delta p_t.$$

Using $\Delta p_t = \mathrm{div}(\nabla p_t) = \mathrm{div}(p_t \nabla \log p_t)$, rewrite the Fokker–Planck equation as a *continuity equation*:

$$\partial_t p_t = -\mathrm{div}\big(p_t\, \tilde{u}_t\big), \qquad \tilde{u}_t(x) = \,?$$

Identify $\tilde{u}_t$ explicitly in terms of u_t, σ_t, and $\nabla \log p_t$.

(b) The continuity equation $\partial_t p_t = -\mathrm{div}(p_t\, \tilde{u}_t)$ corresponds to the ODE $\mathrm{d}X_t = \tilde{u}_t(X_t)\,\mathrm{d}t$. Deduce the **probability flow ODE**:

$$\boxed{\mathrm{d}X_t = \left[u_t(X_t) - \frac{\sigma_t^2}{2}\,\nabla \log p_t(X_t)\right]\mathrm{d}t.}$$

Explain in one sentence why this ODE and the original SDE produce the same marginal distributions p_t despite having different individual trajectories.

(c) Using the conversion formula from Problem 5(b), show that for Gaussian paths the probability flow ODE can be written purely in terms of the vector field u_t^{target}:

$$\mathrm{d}X_t = u_t^{\text{target}}(X_t)\,\mathrm{d}t.$$

Part II: Coding — Flow Matching on 2D Distributions

Implement conditional flow matching from scratch in PyTorch. Train a neural network vector field u_t^θ on a synthetic 2D dataset and use it to generate new samples. Submit a Jupyter notebook (.ipynb) with code, outputs, and brief commentary. All experiments should run on CPU in under 10 minutes.

Task 1: Data and probability paths.

(a) Choose a non-trivial 2D target distribution p_{data} (e.g. two moons, concentric circles, a mixture of 8 Gaussians arranged in a ring, or a Swiss roll). Write a function `sample_data(n)` that returns `n` samples.

(b) Implement the CondOT probability path: given a data sample z and noise $\epsilon \sim \mathcal{N}(0, I_2)$, compute $x_t = tz + (1-t)\epsilon$ and the target vector field $u_t^{\text{target}}(x_t \mid z) = z - \epsilon$. Visualise samples from $p_t(\cdot \mid z)$ for a fixed z at times $t \in \{0, 0.25, 0.5, 0.75, 1\}$.

Task 2: Training.

(a) Define a neural network $u_t^\theta(x)$ that takes as input $(x, t) \in \mathbb{R}^2 \times [0, 1]$ and outputs a vector in $\mathbb{R}^2$. Use an MLP with at least 3 hidden layers (suggested width: 128). Time t should be concatenated to x as an input feature.

(b) Implement Algorithm 3 from the lectures (flow matching training with CondOT path). Train for a sufficient number of iterations (suggested: 10 000–20 000 steps, batch size 256, learning rate 10^{-3} with Adam). Plot the training loss curve.

Task 3: Sampling and evaluation.

(a) Implement Algorithm 1 from the lectures (Euler method sampling). Generate 2 000 samples from the trained model using $n = 100$ Euler steps. Plot the generated samples alongside true samples from p_{data}.

(b) Implement Heun's method (second-order) for sampling. Compare the quality of generated samples using Euler vs. Heun at $n \in \{5, 10, 25, 50, 100\}$ steps. Summarise your findings in a table or figure.

(c) Visualise the learned vector field: plot $u_t^\theta(x)$ as a quiver plot on a grid at times $t \in \{0, 0.25, 0.5, 0.75, 1\}$. Overlay a few ODE trajectories from $t = 0$ to $t = 1$.

Task 4: Extending to diffusion (bonus).

(a) Using your trained vector field u_t^θ and the conversion formula from Problem 5, derive a score network s_t^θ for the CondOT path. Implement Euler–Maruyama sampling (Algorithm 2 from the lectures) with $\sigma_t = 0.1$. Compare the generated samples with the ODE samples from Task 3.

Part III: Paper Review — Discrete Diffusion and Flow Matching for Text

The continuous-state diffusion and flow matching framework studied in Parts I–II operates on $\mathbb{R}^d$. A major open question is how to extend these ideas to **discrete state spaces** (e.g. token sequences), where the data lives on a finite set rather than in Euclidean space. This is the domain of autoregressive LLMs (Weeks 1–3 of this course). Recently, discrete diffusion and flow matching models have emerged as a potential alternative to autoregressive generation for text — notably through the startup Inception Labs (co-founded by S. Ermon), whose diffusion-based language model *Mercury* demonstrates competitive quality with significantly faster inference than autoregressive models.

*Choose **one** paper from the list below. Write a critical review of at most **5 pages** (excluding references). Your review should:*

- *Summarise the paper's main contribution and situate it within the landscape of generative models for discrete data.*
- *Identify the key mathematical idea(s): what replaces the score function, the SDE, and/or the probability path when the state space is discrete?*
- *Critically compare with autoregressive LLMs: what are the architectural, training, and inference trade-offs? Under what conditions might discrete diffusion/flow matching models outperform or complement autoregressive generation?*

1. J. Austin, D.D. Johnson, J. Ho, D. Tarlow, and R. van den Berg. *Structured Denoising Diffusion Models in Discrete State-Spaces* (D3PM). NeurIPS, 2021.
2. A. Lou, C. Meng, and S. Ermon. *Discrete Diffusion Modeling by Estimating the Ratios of the Data Distribution* (SEDD). ICML, 2024.
3. A. Campbell, J. Benton, V. De Bortoli, T. Rainforth, G. Deligiannidis, and A. Doucet. *A Continuous Time Framework for Discrete Denoising Models.* NeurIPS, 2022.
4. J. Shi, K. Han, Z. Wang, A. Doucet, and M. Titsias. *Simplified and Generalized Masked Diffusion for Discrete Data.* NeurIPS, 2024.
5. I. Gat, T. Remez, N. Shaul, F. Kreuk, R.T.Q. Chen, G. Synnaeve, Y. Adi, and Y. Lipman. *Discrete Flow Matching.* NeurIPS, 2024.
6. S.S. Sahoo, M. Arriola, Y. Schiff, A. Gokaslan, E. Marroquin, J.T. Chiu, A. Rush, and V. Kuleshov. *Simple and Effective Masked Diffusion Language Models.* NeurIPS, 2024.